Introduction to Artificial Intelligence

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Contents

1 Introduction to AI **2**
- 1.1 Rational decisions . . . 2
- 1.2 Agents and environments . . . 4
- 1.3 Methods and applications . . . 6
- 1.4 From an application to a computational problem . . . 6

Lecture 1: Introduction to AI

1.1 Rational decisions

- **The goal of the course.** Artificial intelligence can be approached through four goals: making machines think like people, act like people, think rationally, or act rationally. This course focuses on the last: choosing actions that best achieve specified goals. A decision can be rational even if it does not resemble human thinking.
- **The decision rule.** A RATIONAL DECISION maximizes expected UTILITY, a numerical measure of the desirability of an outcome. If h denotes the information available when choosing an action, O is the set of possible outcomes, and $U(o)$ is the utility of outcome o, then for a finite outcome set the rule is
$$a^* \in \arg\max_a \sum_{o \in O} P(o \mid h, a)\, U(o).$$
For each action, weight each possible outcome's utility by its probability, then choose an action with the largest total. Rationality is relative to the goals and information available at decision time: an unlikely bad outcome can occur even when the chosen action had the highest expected utility. The criterion evaluates the decision, not whether its internal reasoning imitates a person.
- **Why uncertainty matters.** The word *expected* matters. An agent usually chooses before it knows which outcome will occur. It should compare the whole distribution of outcomes for each action, not assume that the most likely outcome is certain. Utility also matters: a high probability of success is not enough if the cost of failure is unacceptable.

Example: As a small illustration, suppose a robot can use a safe route worth 6 utility units for sure, or a shortcut worth 10 with probability 0.7 and -5 otherwise. The shortcut's expected utility is $0.7(10) + 0.3(-5) = 5.5$, so the safe route is preferred. If the success probability rises to 0.9, the shortcut's value becomes 8.5 and the decision changes. The same actions can therefore lead to different rational choices when the information changes.

Rationality is evaluated using information available at the time. A decision can be sensible and still end badly. Conversely, a poor decision can get lucky. Judging only the eventual outcome confuses decision quality with chance. The agent's goal is another input: a route that is best for speed may be poor for energy use or safety.

1.1.1 How an objective becomes a decision

A utility function is a modeling choice. Suppose the two routes reach the same destination, but one uses less energy and the other is faster. Merely saying "reach the destination" leaves the choice

unresolved. The utility must express how the agent trades time, energy, and the risk of failing. Probability describes what the agent believes will happen; utility describes how much it values the outcomes. Mixing up these roles leads to mistakes: a rare outcome can still dominate a decision if its cost is large enough.

The route example can be solved for the probability at which the preferred action changes. Let p be the shortcut's success probability. Its expected utility is $10p - 5(1 - p) = 15p - 5$. It is better than the safe route exactly when

$$15p - 5 > 6, \qquad p > \frac{11}{15}.$$

At $p = 11/15$ both actions are optimal under this model. This threshold makes the role of information explicit: improving the estimated success probability from 0.7 to 0.72 does not change the decision, while improving it to 0.8 does. A more accurate prediction is useful for action only insofar as it can affect a consequential choice.

A utility number need not be a monetary amount. It is a way to compare outcomes, including tradeoffs that are difficult to express in a single physical unit. The agent does not choose the utility function by solving the expectation. That function is part of the problem specification. Optimizing the wrong objective very effectively can produce unwanted behavior. For example, maximizing the number of dots collected on the next move can send Pac-Man into a dead end even when surviving to collect more food later is the intended goal.

1.1.2 Human behavior and rational behavior

The four views of AI ask different questions. A system that thinks like a person is evaluated as a model of human mental processes. A system that acts like a person is evaluated by its outward behavior. A system that thinks rationally emphasizes valid reasoning, while a system that acts rationally is evaluated by the quality of its choices. These goals overlap, but one does not automatically establish another. A valid deduction can be useless for the action at hand, and a good action can be chosen without reproducing a human chain of thought.

Human brains motivate the importance of memory and computation, but an AI algorithm need not imitate their physical organization. A route planner can compare thousands of possibilities in a way that a human traveler does not. The useful question for this course is whether the resulting decision fits the stated information and objective, and whether the computation is feasible.

1.1.3 Worked example: the most likely outcome is not the whole decision

Consider an illustrative delivery robot. Route A takes 10 minutes with probability 0.9 and 60 minutes with probability 0.1. Route B always takes 16 minutes. The question is which route the stated objective prefers.

1. **Specify the outcomes.** Route A has two possible arrival times; route B has one. Comparing only 10 with 16 discards the delay event.
2. **Minimize expected time.** The expectations are $0.9(10) + 0.1(60) = 15$ minutes and 16 minutes, so this objective selects A. A bad realized journey does not retroactively make that calculation irrational.
3. **Change the objective explicitly.** Suppose arriving after 20 minutes incurs an additional penalty of 100 utility-cost units, and one minute also costs one unit. The expected costs become $0.9(10) + 0.1(60 + 100) = 25$ for A and 16 for B. Now B is preferred.
4. **Interpret the reversal.** The probabilities did not change. The consequences that matter to the task changed. An agent cannot choose rationally until both its uncertainty and its objective are specified.

This also separates a prediction problem from a decision problem. Predicting the likely arrival time is useful input, but does not by itself determine whether the risk of a large delay is acceptable.

1.2 Agents and environments

An AGENT perceives its environment and acts on it. Sensors deliver PERCEPTS, and actuators carry out actions. The environment then changes and supplies new percepts, so decisions form a continuing interaction. The nature of the observations, available actions, and environment determines which techniques are appropriate.

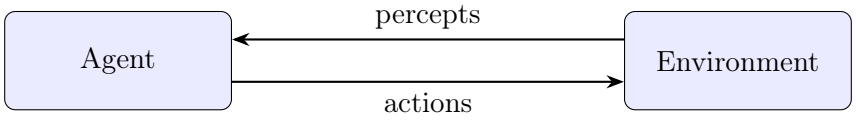


Figure 1: An agent interacts with its environment.

For Pac-Man, the environment is the game, percepts describe what the agent can observe, and actions move Pac-Man through the maze. The agent's objective specifies what counts as a good outcome; seeing the same game does not by itself determine what the agent should prefer.

A REFLEX AGENT chooses from its current percept, possibly using memory, without explicitly evaluating future consequences. A PLANNING AGENT uses a model to consider how candidate actions might change the world and selects a plan in light of its goal. A reflex rule can still be rational when it selects an optimal action for the setting; planning is one way to make decisions, not the definition of rationality.

The distinction between reflex and planning agents concerns *how an action is selected*, not whether the agent stores anything. A reflex agent can maintain an internal estimate of the world and still apply a rule to that estimate. A planning agent goes further: it predicts possible future states and

compares them before choosing.

For example, a rule that turns away from a nearby wall can work well in a simple corridor. At a junction, choosing the branch that eventually reaches food requires comparing consequences beyond the next observation. Planning also has a cost. Looking farther ahead takes computation, and a detailed model can be expensive to use.

An explicit policy stores an action for each relevant situation. Once such a policy has been computed, using it can look like a reflex even if extensive planning was needed to obtain it. Thus the cost of constructing a policy and the cost of executing it are separate questions.

1.2.1 Walking through one decision cycle

Consider Pac-Man approaching a junction. First, sensors provide the current observation: nearby walls, food, and any visible ghosts. The program turns that observation and its memory into an internal description. A reflex rule might choose the first legal direction containing food. A planning program instead creates hypothetical next states for each legal direction, considers what can happen afterward, and compares the resulting outcomes.

The hypothetical moves happen inside the model. The program does not move Pac-Man left in the actual game, undo that action, and then try right. After comparing possibilities, it executes one chosen action. The next observation may reveal that a ghost moved differently from the prediction, so the agent updates its description and makes another decision. This is why planning and replanning belong together: a computed plan is conditional on the model and on what was known when it was computed.

The model can fail even when the search is correct. If it assumes ghosts stay still, finding the best path in that model does not make the actual path safe. If its state forgets which food was collected, it may predict a reward that is no longer available. These are representation errors rather than errors in the rule used to compare candidate plans.

1.2.2 Worked example: one percept can represent different situations

Imagine a vacuum agent that currently observes "this square is clean." There are two squares, left and right. The other square is outside its sensor.

1. **List compatible worlds.** In world w_1 both squares are clean. In world w_2 the current square is clean and the other is dirty. The current percept is identical in the two worlds.
2. **Identify the decision-relevant difference.** With a movement cost, stopping is attractive in w_1, while visiting the other square can be worthwhile in w_2. A reflex rule based only on the current percept cannot distinguish them.
3. **Use remembered information.** If the agent previously visited the other square and cleaned

it, that history can support stopping. If dirt can reappear, the same memory is less decisive; elapsed time and the dirt model may matter.

4. **Choose a sufficient internal description.** The internal state can record location, remembered cleanliness, and uncertainty about changes. It should be updated after both observations and actions. It is a summary for future decisions, rather than a complete recording of every past event.

The example shows why an agent's input at one instant is not automatically a complete state. Observability is a property of what the sensors reveal about the task, not of how much computation the agent performs afterward.

1.3 Methods and applications

AI systems combine computation and data. Search and planning compare possible actions; constraint satisfaction finds assignments that obey specified rules; adversarial and uncertain settings require reasoning about other agents or uncertain outcomes. Probabilistic models represent uncertainty, decision theory connects uncertainty to utility, and machine learning uses data to improve predictions or decisions.

The same principles appear in different applications:

- **Language:** speech recognition maps audio to text, speech synthesis maps text to audio, and language systems support translation, question answering, and text classification.
- **Vision:** perception turns pixels into information, such as object identities, scene labels, or three-dimensional structure.
- **Robotics:** perception, planning, and control connect decisions to physical actions. The real world can differ from a simulator.
- **Games and logic:** game-playing systems reason over moves; logical systems use deduction, constraint satisfaction, and satisfiability to solve structured problems.

The field has used both knowledge-based and statistical approaches. Recent systems in the 2018 course draw on large datasets, computation, and neural networks; the appropriate representation and objective still depend on the task.

1.4 From an application to a computational problem

To build an agent, first make the decision problem explicit. What can it observe? Which actions can it take? What changes after an action? What outcomes does the objective reward? These choices determine what the agent must represent and what it can safely ignore.

In a maze, a route planner may need only locations and legal moves. If ghosts affect survival, their positions or a belief about their positions become relevant too. If the objective is to collect all food, remembering only the current location is insufficient because the remaining food depends on the past. Choosing a representation is therefore part of solving the problem, not merely a coding detail.

The major methods fit different gaps in this description. Search finds a good sequence when the model is known. Game search accounts for other decision makers. Probability represents unknown quantities. Learning estimates models or policies from examples and experience. A complete application often uses several of these methods together.

1.4.1 How computation and data complement each other

The distinction between computation and data is useful when choosing a method. If all roads and their travel costs are known, the remaining task may be to search efficiently for a route. If traffic costs are unknown, data are needed to estimate them before or during planning. More search cannot by itself reveal an unknown road closure, while more observations do not by themselves identify the cheapest route through a large network. A system needs a representation that allows the estimates and the planner to work together.

The field's historical changes reflect different ways of supplying this information. Early programs demonstrated search and symbolic reasoning. Knowledge-based systems emphasized explicit rules provided by people. Statistical approaches emphasized learning from observations and reasoning with uncertainty. The neural-network successes discussed in the 2018 course combine large datasets with substantial computation. These approaches do not remove the need to specify a task, choose a representation, and evaluate whether the decisions meet the objective.

For language, the observations may be audio or written words, while the output might be a transcript, translation, or label. For vision, pixels must be converted into information such as objects and geometry. For robotics, those estimates feed actions that change the physical world. For games, other players change which futures are plausible. The general techniques are valuable because these different applications share recurring structures: search over alternatives, uncertainty about hidden quantities, and learning from incomplete experience.

Source. Pieter Abbeel and Dan Klein, Introduction to Artificial Intelligence, Fall 2018, Lecture 1 and Note 1.